LFCPNX-100-9CBG256I pinout for the MIPI CSI-2 to 2.5GbE bridge

Device CertusPro-NX, 100K LC, speed grade -9, CBG256 (fcCSP, 14 x 14 mm), industrial **Proposal** revision 3

Pinout check 5 November 2026

Context

The design receives a 2-lane MIPI CSI-2 stream at 1.5 Gbps per lane, writes every frame to an external x16 LPDDR4 buffer at 1066 Mbps, then reads it back and sends it as UDP over a single 2.5GbE link through an external 2.5GBASE-T PHY. The receiver configuration is fixed at compile time and there is no soft CPU: the sensor and the PHY are set up from fixed register lists.

The pinout is the hard part of this design. The CBG256 package has only 165 user I/O, and LPDDR4 and the soft MIPI D-PHY both need the high-performance banks (3, 4 and 5), but at different I/O voltages: LPDDR4 requires 1.1 V and the MIPI low-power mode requires 1.2 V. The plan below gives each interface its own banks. LPDDR4 reuses the x16 part of the placement Lattice uses on the CertusPro-NX Versa Board (FPGA-RD-02278), and every DQS group it needs is bonded out in this package.

PIN PLAN

Accepted by Radiant 2026.1: all 48 constrained ports placed on the planned balls

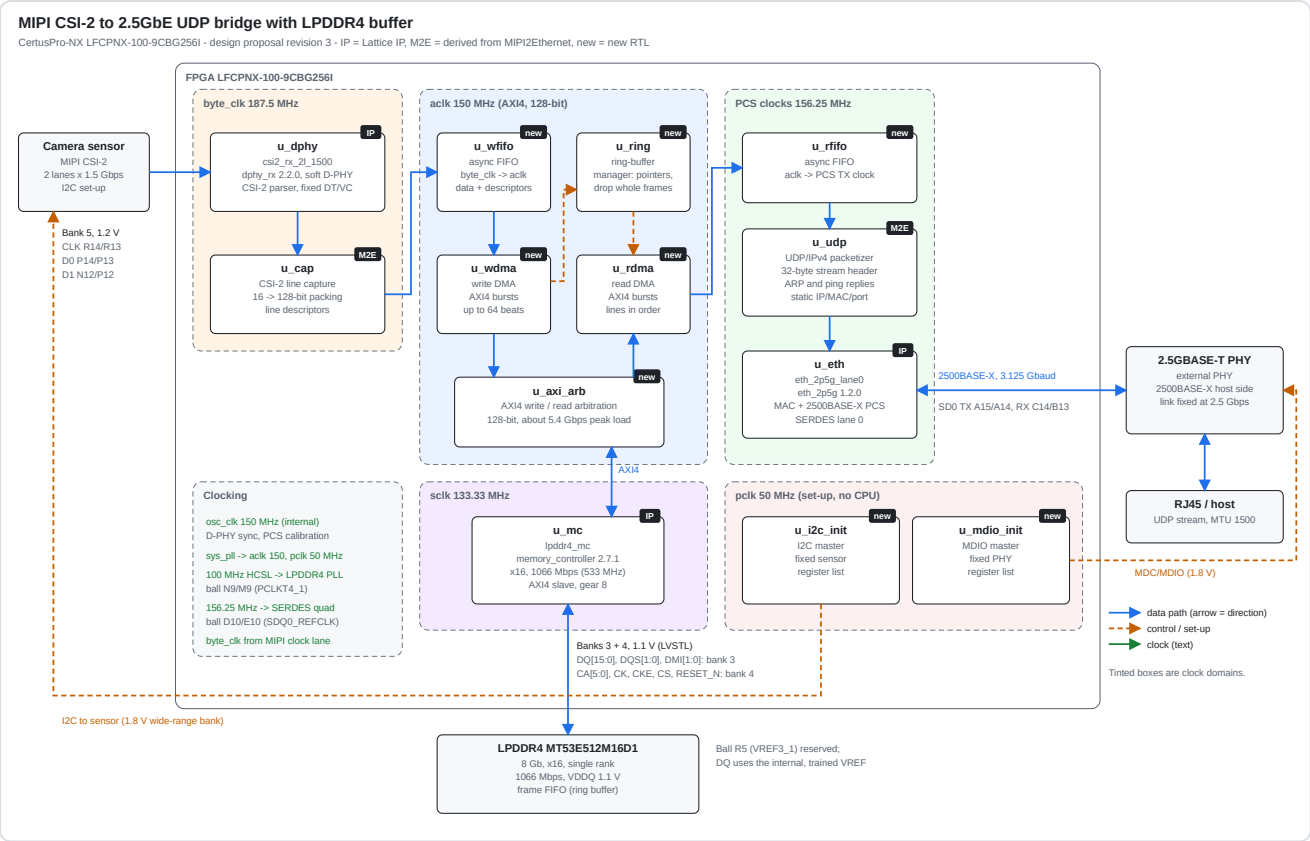
TIMING (PINOUT CHECK BUILD)

Met at 100 °C and -40 °C, setup and hold

HARDWARE

Not tested: no board with this package yet

Block diagram



Proposed bridge architecture. Tinted boxes are clock domains. IP = Lattice IP, M2E = derived from the earlier MIPI2Ethernet project, new = new RTL.

Package resources

RESOURCE	LFCPNX-100 IN CBG256
User I/O	165: 84 high-performance (banks 3, 4, 5), 75 wide-range (banks 0, 1, 2, 6, 7), 6 ADC inputs
SERDES	4 lanes, SD0 to SD3, in one quad
HP bank structure	Whole 12-pin DQS groups. Bank 3: groups 114, 150. Bank 4: groups 54, 102. Bank 5: groups 6, 30, 42.
MIPI rate limit	1500 Mbps per lane needs a flip-chip package and the -9 grade; CBG256 is flip-chip. Wire-bond packages stop at 1250 Mbps.

Bank voltages

BANK	VCCIO	USE
3	`1.1 V`	LPDDR4 data (DQ, DQS, DMI)
4	`1.1 V`	LPDDR4 command and address, clock, reset, 100 MHz reference clock
5	`1.2 V`	MIPI CSI-2 (30 pins spare, 1.2 V signals only)
0, 1, 2, 6, 7	`1.8 V`	SPI configuration flash (bank 0), JTAG (bank 1), sensor I2C, PHY MDC/MDIO, resets, status
VCCAUXH3/4/5	`1.8 V`	Auxiliary supplies of the HP banks

Interface ball assignments

LPDDR4 x16, MT53E512M16D1 (8 Gb), 1066 Mbps

SIGNALS	BANK	DQS GROUP	BALLS
Byte 0: DQ[7:0]	3	150	`N3 P2 M3 N2 M4 L4 M2 L2`
DMI0, DQS0 (t/c)	3	150	`M1; P1/N1`
Byte 1: DQ[15:8]	3	114	`P4 P5 M5 N4 R4 R3 R2 P3`
DMI1, DQS1 (t/c)	3	114	`N5; T3/T4`
CA[5:0]	4	54	`K10 L10 P10 R10 T10 T9`
CKE, CS	4	54	`R9, P9`
Reference clock 100 MHz (HCSL), PCLKT4_1	4	54	`N9/M9`
CK (t/c)	4	102	`P8/R8`
RESET_N	4	102	`L7`

MIPI CSI-2, 2 lanes x 1.5 Gbps

LANE	BANK / GROUP	BALLS P/N
Clock (PCLKT5_2)	5 / 42	R14/R13
Data 0	5 / 42	P14/P13
Data 1	5 / 42	N12/P12

2500BASE-X to the PHY, SERDES lane 0

SIGNAL	BLOCK	BALLS P/N
TX	SD0	A15/A14
RX	SD0	C14/B13
Ref. clock 156.25 MHz	SDQ0	D10/E10

Pin budget

GROUP	AVAILABLE	USED	SPARE
Bank 3 1.1 V	24	22, plus R5 reserved	1
Bank 4 1.1 V	24	13	11
Bank 5 1.2 V	36	6	30 (1.2 V only)
Wide-range banks 1.8 V	75	About 25 (configuration, JTAG, control)	About 50
SERDES lanes	4	1	3

Notes for the board designer

- Supply VCCIO3 and VCCIO4 (balls R1, T2, R7, T11) from a 1.1 V rail, which is also the LPDDR4 VDDQ. Supply VCCIO5 (R16, T15) from a separate 1.2 V rail.
- Drive the LPDDR4 reference clock from a 100 MHz HCSL oscillator, DC-coupled with 49.9 Ω to ground (FPGA-TN-02244, Section 6.4.2.1). An LVDS oscillator would need AC coupling and a DC bias network.
- Drive the SERDES quad reference clock from a 156.25 MHz differential oscillator.
- AC-couple the 2500BASE-X TX and RX pairs. Any lane SD0 to SD3 can be used to ease routing to the PHY; changing lanes is a single IP setting.
- Route the MIPI pairs to MIPI D-PHY impedance rules. The FPGA's internal 100 Ω termination is used, so no external termination resistors are needed.
- Both MIPI (1500 Mbps) and LPDDR4 (1066 Mbps) run at the device's -9 edge-clock limits, so signal integrity on these interfaces has no margin to give away.

Sources:

- FPGA-SC-02022 rev 1.0: CertusPro-NX 100K pinout (CBG256 column)
- FPGA-DS-02086 rev 2.5: CertusPro-NX Family Data Sheet
- FPGA-TN-02244 rev 1.6: CertusPro-NX High-Speed I/O Interface
- FPGA-TN-02090 rev 1.4: MIPI D-PHY Bandwidth Matrix and Implementation
- FPGA-IPUG-02127 rev 2.2: LPDDR4 Memory Controller IP Core for Nexus Devices
- FPGA-RD-02278 rev B: CertusPro-NX RISC-V and LPDDR4 Memory Controller Reference Design
- Radiant 2026.1 pinout check build of project `MipiEthernetLpddr4`

Standards: MIPI CSI-2 and D-PHY v1.2; JEDEC JESD209-4 (LPDDR4); IEEE 802.3bz (2.5GBASE-T, external PHY) and IEEE 802.3 Clause 127 (2.5GBASE-X PCS).